

Interlibrary loan in Türkiye: interlibrary loan tracking system (KITS)

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Abstract

Purpose – This study aims to evaluate the current situation of the interlibrary loan in Türkiye using interlibrary loan tracking system raw data (2008–2019). Data was analyzed geographically to reveal the effect of geographical distance on the collaboration and thematically to identify the prominent areas/topics in the requests. In addition, the requests were evaluated in terms of the type and age of the university, the size of the collection and the number of library users.

Design/methodology/approach – The effect of geographical distance on the collaboration was revealed using VOSviewer. The place numbers and titles of the requests were used for thematic analysis, which was performed using Flourish and VOSviewer. Statistical tests were conducted with SPSS to investigate the effects of factors such as university age and type.

Findings – Geographic analysis revealed that the prominent regions are Marmara, Central Anatolia and Aegean Regions, respectively. The fact that these regions host Türkiye's largest cities with the highest number of universities has been effective in achieving this result. The results of the thematic analysis are in accordance with the literature. Almost half of the requests are from Social Sciences, Language and Literature, History and Law. While the effects of collection size and the number of library users have relatively insignificant effects on requests, age and university type were found to be more relevant.

Originality/value – Interlibrary loan data of this size has not been the subject of an evaluation before in Türkiye. Hence, the results obtained have the potential to influence relevant policies and decisions on the subject. On the other hand, to the best of the authors' knowledge, bibliographic mapping was used for the first time on interlibrary loan data.

Keywords Interlibrary cooperation, Interlibrary loan, Türkiye, KITS, Interlibrary lending, Resource sharing

Paper type Research paper

1. Introduction

The main function of libraries is to provide services to users through collections. Libraries strive to cover the changing needs of their users by constantly updating their collections. However, due to limited budgets and the increasing number of materials available, libraries are often unable to fulfill all the requests for new resources from their users. Libraries, which are responsible for meeting the information needs of the users, cooperate with other libraries and information centers for economical and fast material sharing to overcome this situation. American Library Association (ALA) Adult Services Division emphasized the importance of cooperative resource sharing, stating that the library can meet the information needs of users with its own collection and local, regional or national collaborations for resource sharing among libraries (Bustos, 1993, p. 25). Although interlibrary cooperation cannot completely solve problems in an environment where many different external factors are in question, it is one of the most effective ways to alleviate the difficulties experienced in providing materials.

Libraries can cooperate for cataloging, collection development and interlibrary loan (ILL) (Britannica, 2022). Since the early 1900s, cooperative cataloging has allowed for contributions to the bibliographic information of materials using a set of standards. OCLC WorldCat stands out as a prominent example in this regard, with Ohio University's Alden Library being the first to use WorldCat to catalog a book online in 1971, marking the launch of the OCLC Online Union Catalog and Shared Cataloging System (Seaman, 2016). During the same period, the Library of Congress (LC) also began its cooperative cataloging efforts (Library of Congress, 2023). The national cooperative cataloging system of Türkiye, TO-KAT, is another example of cooperative cataloging at the national level (TO-KAT, 2022). Cooperative collection development, which emerged in the 1940s, prevents the same material from being purchased by different libraries and provides benefits both in terms of storage and economy. Another form of cooperation between libraries is ILL. ALA defines ILL as the “process by which a library requests

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material from, or supplies material to, another library.” In this definition, material refers to “books, audiovisual materials, and other returnable items as well as copies of journal articles, book chapters, excerpts, and other non-returnable items” (ALA RUSA, 2020). The concept of ILL was first introduced by Samuel Swett Green, who worked at the Worcester Library, in 1876. Green sent a letter to the editor of the *Library Journal*, stating that libraries could make an agreement to lend each other books, albeit for a short time. In the letter, he mentioned that accessing the book through another library would be easier in every aspect, rather than waiting for the purchase process of the book, and that libraries would not refuse when such a request came, and they would agree to help each other (Swett Green, 1876, p. 15).

The University of California Library, LC and ALA took the first concrete steps in terms of ILL. The University of California Library announced in 1898 that it could lend the resources in its collection to other libraries (Weber, 1976). In the following years, the LC developed an ILL policy to lend its materials internationally (Stuart-Stubbs, 1975). ALA, on the other hand, created the national ILL code in 1917, which includes the responsibilities of the two libraries to each other while collaborating, and in 1919, it determined and published the ILL rules (ALA RUSA, 2020; Weible and Janke, 2011). Founded in 1927, IFLA initiated the international ILL process in 1934. In 1939, the ILL form and codes created by IFLA were accepted by the cooperating countries (Ryward, 1994, p. 344).

In 1979, significant enhancements were introduced to the ILL and document supply systems, including the introduction of messaging capabilities for the first time through the system (Frederiksen *et al.*, 2012, p. 23). On April 1, 1979, the OCLC resource sharing network processed its first ILL request through its systems, and during the same year, the network successfully handled a total of 565,680 ILL transactions (King, 2019). These advancements played a crucial role in automating the ILL process, thus streamlining the process of borrowing and sharing resources among libraries. Further developments continued to be made in subsequent years to further improve the system. Through DOCLINE, the document supply system of the National Library of Medicine, which was established in 1985 to deliver medical collection to people in need, 65 million resource requests were met within the scope of ILL service until 2020 (Theisen, 2020). Then, the ARIEL document delivery system was developed by the Research Libraries Group in 1990 (Landes, 1997). In the following years, many similar platforms such as Clio, RapidILL and ILLiad have emerged (Clio, 2021; OCLC ILLiad, 2021; RapidILL, 2021). An important development in this process is the publication of the ISO ILL standard (ISO 10160:1997) (Government of Canada, 2004).

One of the most important recent examples of interlibrary collaborations intensified for research and studies on COVID-19 is IFLA's platform called “Resource Sharing during Covid-19 (RSCVD)”. The HERMES project, funded by the European Union through the Erasmus+ program, has been implemented to ensure that the RSCVD platform has a longer-lasting impact and turns it into a multifaceted action (HERMES, 2021). In unpredictable situations such as economic crises, disasters and pandemics, sharing information with the ILL is even more important. During the COVID-19 pandemic, the importance of information sharing has been better understood, especially in the

context of vaccine and drug development studies. On the other hand, the economic impact of the COVID-19 pandemic has affected Türkiye as well as many other countries. University libraries, which pay for most of their resources in foreign currency, are experiencing significant budgetary difficulties due to increasing exchange rates and are faced with the problem of unsubscribing from many important databases. Using ILL effectively has been suggested as a solution to the budget problems experienced by university libraries in Türkiye during the 1977–1982 economic crises (Tonta, 1987). This recommendation, which was made 35 years ago, is still valid today. The first step for the effective use of the ILL is a detailed analysis of existing system data. By offering valuable insights into the ILL process, the data collected provides a controlled follow-up to the process and can effectively inform how libraries respond to evolving user needs.

There are currently two ILL systems in Türkiye. The first one is the Interlibrary Loan Tracking System (KITS) developed by the Anatolian University Libraries Consortium of Türkiye (ANKOS) in 2008, and the other one is Türkiye Document Supply and Lending System (TÜBESS) developed by

the Turkish Academic Network and Information Center (ULAKBİM) in 2011. However, the history of interlibrary lending in Türkiye dates back to the 1960s. Turkish Scientific and Technical Documentation Center, established in 1966, became the institution that created a structure related to the subject and developed cooperation processes in Türkiye (Toplu, 2009), and activities in this direction continued with the Council of Higher Education (YÖK) Documentation Center (YÖKDOK) established in 1983 (Tuncer, 1988). With the protocol signed between YÖK and TÜBİTAK in 1996, these activities were transferred from YÖKDOK to TÜBİTAK ULAKBİM (National Academic Network and Information Center) (Yörü, 2012). ULAKBİM document supply service received requests by the postal method in 1996–2000, but later it started to be received electronically with the document supply system. Within the scope of the OBES (Common Document Supply System) project, which is the TÜBİTAK ULAKBİM document supply system, a protocol was signed with METU, Hacettepe and Gazi Universities in 1999, and it aimed to deliver the requests to the users quickly and cost-effectively by using the collections of member libraries (Cebeci, 2003; Toplu, 2009). Although requests were made electronically as of 2007, the electronic sending of resources did not start in the same period (Toplu, 2009). TÜBESS, which is one of the current ILL systems in Türkiye, was developed by ULAKBİM within the scope of a protocol signed with the Ministry of Culture and Tourism. The protocol was made in accordance with the e-Integrated Library System in the 38th article of the State Planning Organization Information Society Strategy (2006–2010) and the Additional Action Plan (ULAKBİM TÜBESS, 2021). KITS, another existing interlibrary lending system, started to develop in 2006 by ANKOS, which was established in 2000 and enabled libraries to provide resources to their institutions at a lower cost (ANKOS KITS, 2023).

This study analyzes KITS raw data to reveal the current situation in Türkiye in terms of the ILL. The research questions to be answered in the study are as follows:

- RQ1.* Which topics, resource types and formats (electronic/print) stand out?
- RQ2.* Which universities benefit the most? Do the collections of universities and the number of library users affect the number of requests made/received?
- RQ3.* Which universities cooperate the most? Does the geographical distance between universities have an effect on these collaborations?
- RQ4.* Does the use of KITS vary according to the type (state/foundation) and age of the university?

2. Literature review

Reviewing the existing literature, besides the studies that analyze ILL systems data similar to this study, there exist studies that explore various aspects of ILL. These include the studies focused on the development of ILL systems, evaluations of ILL systems and assessments of the service quality provided by ILL systems.

2.1 Development and evaluation of interlibrary loan systems

Several studies have explored the feasibility of national ILL systems in different countries. For example, Siddiqui (1995) conducted a study in Saudi Arabia to measure the feasibility of a national ILL system, which revealed factors such as insufficient collection, budget, communication, personnel, planning and organization affecting cooperation. Another study by Igwe (2010) in Nigeria highlighted the failure of related projects due to coordination issues, lack of infrastructure, foreign technology dependency and professional inadequacy of librarians. Similarly, Owolabi *et al.* (2011) found that despite the importance of resource sharing in Nigerian libraries, inadequate resource sharing levels and security concerns hindered effective implementation. Recommendations included raising librarian awareness and allocating resources in university budgets. In Wuhan, China, Xing *et al.* (2020) conducted a feasibility study that demonstrated the advantages of electronic resource sharing, proposing the use of cloud computing for cost-effective and efficient distribution. Moreover, regional consortiums and cooperation were deemed suitable for resource sharing due to the unique collections held by university libraries.

In developing ILL software, Li and Yang (2018) identified essential layers for the technological infrastructure as service, transaction, physical resource and virtual management. Studies have primarily focused on evaluating specific systems, such as the Borrow Direct ILL system (Collins, 2015), UnityUK and Fab libraries (Froud, 2016). Consortia have also been examined, highlighting the cost benefits of centralized systems (Jong, 2016) and the request coverage ratio of the J-Gate system in India (Panda *et al.*, 2016). Studies note a decrease in requests over time due to various factors, including expanding library collections, open access, licensing restrictions, technological advancements and limited librarians' limited ability to access information (Goolabsingh, 2019; Panda *et al.*, 2016; Stapel, 2016).

2.2 Assessment of interlibrary loan service quality

The importance of a dedicated staff responsible for ILL services and user access to relevant Web pages for information retrieval has been emphasized (Panda *et al.*, 2016). Evaluating service quality through surveys and tests provides valuable insights. A questionnaire-based evaluation of ILL services at New York College Library yielded positive feedback from users (Landes, 1997). The Jiangsu Academic Library and Information System in China concluded that developing small consortia would be beneficial for service quality (Xi *et al.*, 2016). In China, the LIBQUAL scale was used to assess service quality in Chinese digital libraries, highlighting the need for improved research skills among librarians and increased awareness of ILL services (Xi *et al.*, 2016). A Canadian study used daily notes from users visiting public libraries, revealing low service quality and extended borrowing periods for ILL services of the public libraries (Hebert, 1993). Similarly, a study at Texas A&M University found a low utilization rate of free document delivery services and recommended raising awareness about ILL services (Yang *et al.*, 2019). Process improvement efforts at the University of Arizona involved personnel training and monthly meetings to accelerate turnaround times (Voyles *et al.*, 2008). A comprehensive evaluation of the service quality of ALA STARS highlighted an increase in the use of ILL services and the evolving tools used over time (Munson *et al.*, 2016).

2.3 Studies evaluating interlibrary loan system data

The analysis of request data at a geographical level, like in this study, is a research area that lacks sufficient studies in the literature. An example of such a study is conducted by Thompson *et al.* (2019), where they examined the sizes of resource requests at the country level for Latin American countries using data from the Big Ten Academic Alliance. However, it should be noted that existing studies using request data tend to focus more on the subjects that are highly requested.

In Israel, it was found that social sciences, humanities, education and health sciences came to the fore in resource requests, and it was concluded that while there was a significant relationship between resource requests and the number of university collections, there was no relationship between the resource request and the year of establishment and the number of collections. On the other hand, the high number of requests from some libraries is due to the policy of these libraries to lend all kinds of resources. In light of the analysis results obtained, it is recommended to increase the number of personnel responsible for ILL and to develop appropriate policies to meet the requests (Porat, 2003). Another similar study (Aguilar, 1984) was conducted at the University of Illinois, and it was aimed to determine whether the resource request varies according to the subject. The subjects of the resources are determined according to the subject numbers of The Library of Congress Classification, The Dewey Decimal Classification (DDC) and National Library of Medicine Classification. It was seen that there were more than one copy of resources from the most used subject classes. One of the other remarkable findings is that less in-house borrowed materials are also less requested through the ILL system. In another study (Knievel *et al.*, 2017) based on three-year circulation and ILL data of the University of Colorado with a focus on thematic analysis, the subjects of the most used resources in circulation and ILL were found to

be common (language, literature, science, history and auxiliary sciences, business and economics, engineering and technology, philosophy). In the study, it was concluded that the findings should be used in collection management.

Studies examining the increase or decrease in requests over time and trying to find the reason for this also have an important place in the literature. In one of these studies (Kim *et al.*, 2019), requests for cancellation were studied. As a result of the study, it has been determined that the discovery of resources through discovery tools is effective in the cancellation of requests; on the other hand, it increases the ILL requests significantly, and the correct use of the discovery tool reduces the workload of the librarian. The main reason why the requests vary according to the resource type is the ILL policies. For example, Aguilar (1984) noticed that although textbooks are widely used in-house, they are less requested through ILL systems. ALA does not have a clear statement regarding the lending of textbooks within the scope of ILL. Recent studies to produce solutions for this attract attention (McNeil, 2017). There is another factor that can affect how many requests are made, which is how much people are charged. For instance, Porat *et al.* (2020) found that when ILL fees were removed, there was a 69% increase in requests for national and international resources in 2020 compared to the previous year.

2.4 Studies on interlibrary loan in Türkiye

Before the development of KITS and TUBESS systems, there were several national studies addressing the issue of ILL. In a thesis prepared in 1991 (Demir, 1991), ILL was discussed in the context of data collected through observation in the İstanbul University Library. As emphasized in a study presenting the situation 20 years ago, “interlibrary cooperation requires commitment, mutual understanding, consensus and patience” and it provides significant benefits to the budget (Tonta, 1999). In a study based on the document supply data of ULAKBİM, which is an important institution on the subject in Türkiye, the contribution of document supply to library budgets was evaluated (Ünal, 2002). The institutions that requested the most resources were found to be İstanbul, Ege and Yüzüncü Yıl Universities. In another study (Alkış and Yılmaz, 2008) in which interlibrary lending and document supply were evaluated through Bilkent University, the effect of the institutionalized interlibrary lending process on service quality and cooperation was mentioned, and it was stated as very necessary to expand collaborations in terms of the future of libraries, especially in countries that have the potential to experience resource/budget problems. An important suggestion of the study is that institutions share their ILL and document supply rules more clearly and regularly. Additionally, there is a study by Toplu (2009) that examines the international aspects of document supply using data from the ULAKBİM Common Document Supply System (OBES). This research provides an analysis of the issues encountered in this context and proposes potential solutions to address them.

In 2010, before the current study, the ANKOS interlibrary collaboration group evaluated the KITS data. The evaluation found that the most frequently requested institutions were METU, İstanbul Bilgi University and İstanbul Technical University, in that order. Additionally, the institutions that submitted the highest number of requests were Sabancı, Koç and Yeditepe Universities. The request response time was

determined as one day on average for 64.4% of the requests (Çimen *et al.*, 2010). After this study, a questionnaire was applied to the personnel responsible for ILL and document supply in university libraries. According to the survey answered by 100 institutions, 66% of the requests are answered within 1–3 days, and geographical proximity is the most important consideration in the selection of the institution to be requested. In the study, in which it was learned that KITS is generally used for domestic requests, it was also determined that the articles in the license agreements negatively affect the lending process and that different license models should be developed. The problems experienced in the requests for rare works, theses and e-books due to the unclear institutional conditions are also mentioned. In addition, it was stated that there should be a national integrated system such as OCLC (Celikbas and Ekingen Flores Mamondi, 2016).

There have also been studies in the national literature in which ILL statistics of universities were evaluated. In one of them (Güran and Kaya, 2017), the ILL statistics of Atılım University for the years 2009–2016 were examined, and it was determined that the sources in the fields of law, social sciences and literature were mostly lent and borrowed. It has been understood that the subjects of the borrowed sources and the subjects of the sources included in the collection overlap, and it has been concluded that ILL is beneficial in terms of collection development. In addition, it is emphasized that more awareness should be created about interlibrary lending to encourage users. Another example of examining the data within the university is the study of Bilgi University’s on-campus resource sharing (Cuhadar *et al.*, 2019). It was found that mostly books were sent between campuses. Thematically, resources are requested in the fields of language and literature and social sciences. It has been determined that resource sharing in university libraries in Türkiye continued in 31% of institutions during the COVID-19 pandemic period. One of the most current problems in Türkiye regarding resource sharing is insufficient policies for e-resource sharing (Çimen *et al.*, 2020).

3. Methodology

3.1 Data collection

In this study, it is analyzed whether there is a relationship between the use of KITS and the type of universities (state, foundation), foundation years (age), collection size and number of users. For this purpose, first of all, the list of universities using KITS was accessed from the KITS website (<https://kits.ankos.gen.tr/>). For the library collections and the number of users of these universities, university library statistics of 2019 obtained from YÖK were used. The type of universities, their city and geographical region information are also accessed from the universities Web page of YÖK (www.yok.gov.tr/universiteler/universitelerimiz). In the study, the foundation years of the universities were also needed to calculate the ages of the universities in 2019, and this information was obtained from the universities’ own websites to be a primary source. Note that, in Türkiye, there are two types of universities: state universities and foundation universities. State universities are publicly funded, whereas foundation universities are nonprofit private institutions. Foundation universities are administered and funded by private

entities (foundations) instead of the government (Study in Türkiye, 2019).

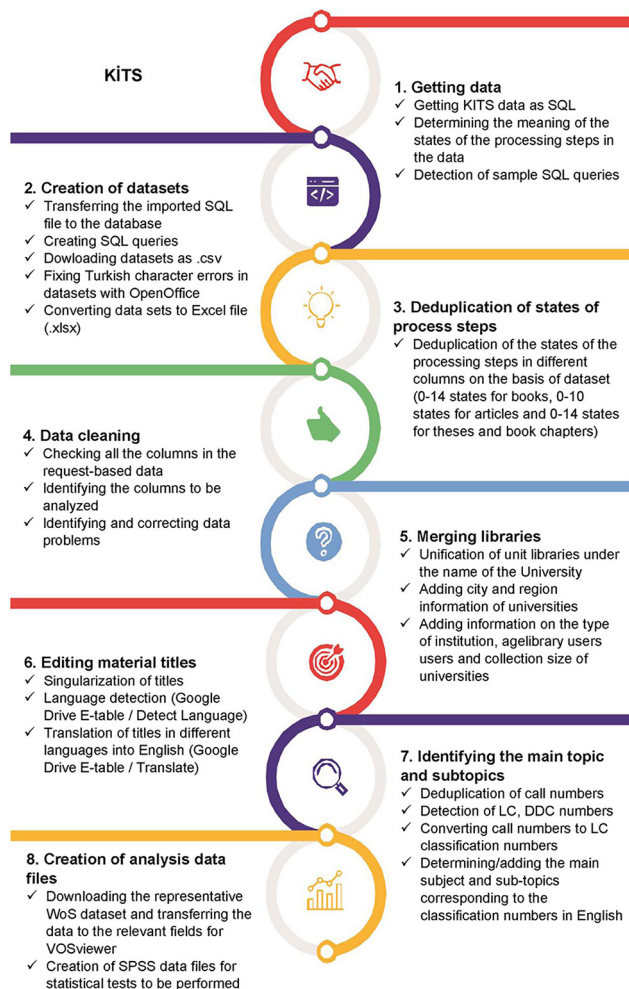
ANKOS provided the KITS data, which included all records from the start of the system until June 2020, in SQL format. After analyzing the data, three separate sets were created: one for books, another for articles and a final one for theses and book chapters.

3.2 Preparing data for analysis

KITS raw data (2008–2019) for the universities providing education in Türkiye as of 2019 were analyzed within the scope of this study. All other data were excluded (for example, nonuniversity institutions, closed universities). In addition, all data that lacks the requesting (borrower – which will borrow) and/or requested (lender – which will lend) university information were also excluded. Figure 1 presents the process of preparing the KITS data for analysis. The data kept in Excel for each of the three data sets and the data used in the study are presented in Appendix 1.

Table 1 shows the data size in each of the three data sets before and after the data cleaning process.

Figure 1 Steps of preparing KITS data for analysis



Source: By authors' Çölden Akgül & Doğan

Note that since the call numbers are only available for the first data set (books), thematic analysis is conducted in the first data set, which is approximately 83% of all data. LC main headings corresponding to DDC main headings and their distribution in the data set are presented in Appendix 2.

3.3 Data analysis

One of the aims of this study is to reveal whether geographical proximity has an effect on ILL. For this purpose, cooperation between regions, cities and universities was examined. VOSviewer (see www.vosviewer.com/), a bibliometric visualization tool, was used to perform geographical analysis. KITS data set has been brought to Web of Science (WoS) format so that it can be run on VOSviewer. For this purpose, a representative data set was downloaded from WoS. Requested and requesting region, city and university information has been added to the address (C1) field of the WoS data set. Three VOSviewer maps were created to show the cooperation between regions, cities and universities.

The study also aimed to identify the areas and subjects that use ILL services the most. For this thematic analysis, the call numbers were used to determine the main and subheadings of the sources. The main and subheadings were visualized using a radial tree/map created via Flourish visualization software (<https://flourish.studio/>) and frequency tables based on requests. For a more detailed thematic analysis, material titles were used. VOSviewer software was used to visualize the words/phrases in all material titles. Since the titles are available for all three KITS data sets, the VOSviewer map for titles is based on all data, which includes 1,536 words/phrases that are common in 40 or more titles.

The number of requested and requesting resources can vary whether the university is a state or foundation university, how old it is, and the number of collections and users of its library. The Mann–Whitney *U* test was used to determine whether the number of requested and requesting resources differ by university type (state/foundation). The Spearman correlation test was used for the effect of university age, collection size and the number of users. In addition to the analysis based on age, the Mann–Whitney *U* test was conducted to examine the difference between the requests of universities younger than 20 and aged 20 and older. All statistical tests were performed with SPSS (95% confidence level), and effect sizes for positive tests were also calculated manually using Cohen's formula (Cohen, 2013a, 2013b).

4. Findings and results

4.1 Geographical findings

There are seven geographical regions, 81 cities and 208 universities in Türkiye. In total, 80 cities and 161 universities from seven geographical regions benefited from KITS. The numbers of cities and universities from each region that cooperate through KITS are as follows:

- 1 Marmara Region: 61 universities from 11 cities;
- 2 Central Anatolia Region: 31 universities from 13 cities;
- 3 Aegean Region: 13 universities from eight cities;
- 4 Mediterranean Region: 13 universities from eight cities;
- 5 Eastern Anatolia Region: 15 universities from 14 cities;
- 6 Black Sea Region: 18 universities from 18 cities; and
- 7 Southeastern Anatolia Region: 10 universities from eight cities.

Table 1 Requests numbers in three KITS data sets before and after data cleaning

Data sets	Material type	Number of requests			
		Before data cleaning		After data cleaning	
		<i>n</i>	%	<i>n</i>	%
Data set 1	Books	172,632	83.0	150,466	82.6
Data set 2	Articles	27,631	13.3	24,892	13.7
Data set 3	Thesis, Book chapters	7,625	3.7	6,675	3.7
	Total	207,888	100.0	182,033	100.0

Source: By authors' Çölden Akgül and Doğan

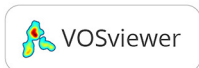
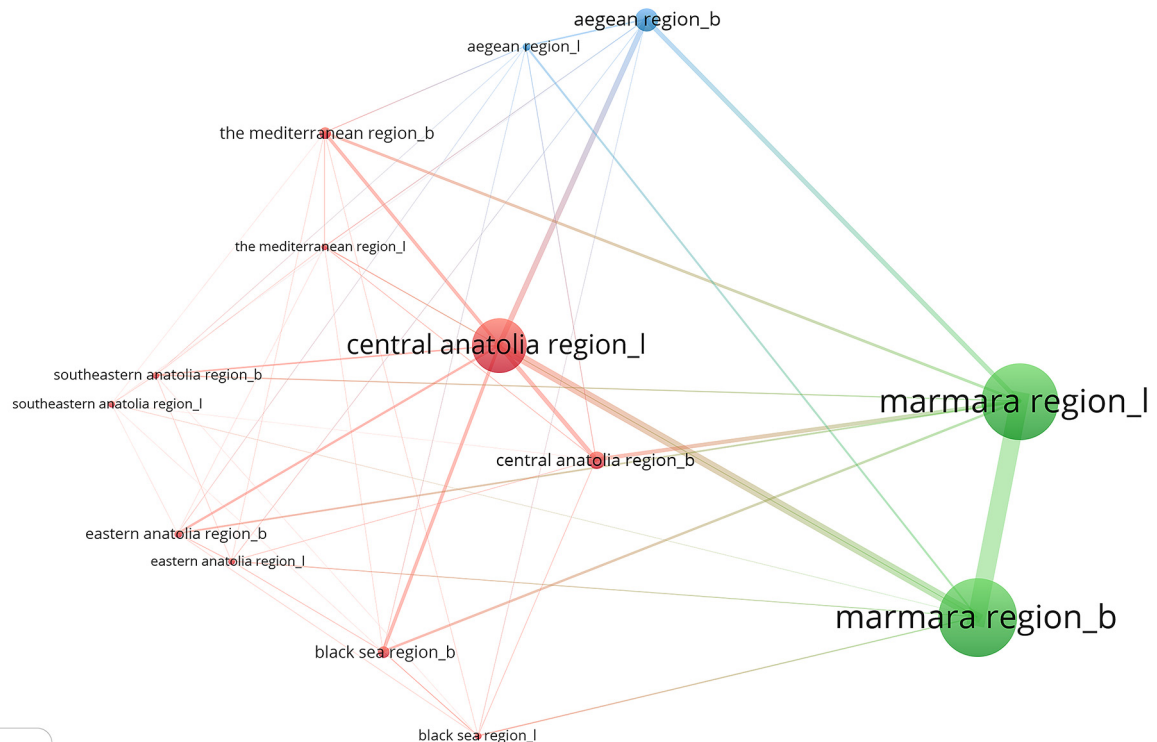
Figure 2 shows the regional cooperation network. Regions as requesting resources are shown with -b (borrower) after the region name, and regions as resources requested are shown with -l (lender). The network is based on the 182,033 new request data of 161 universities.

Different colors in Figure 2 represent regions with the most collaborating university groups. The proximity of the regions to each other means that they cooperate more. The size of a node representing a region in the network indicates the level of resource requests either from or to other regions within the network. The thicker the connections between regions, the greater the cooperation. There are three groups of regions represented by three different colors; green, red and blue. Although the cooperation between the same colored regions are more intense, there is also cooperation with the regions in

other-colored groups. Note that, while interpreting the network, the number of universities in the regions is a factor that should be considered.

The most dominant region in the network is the Marmara Region. The Marmara Region is the cluster with the highest number of requests. While the number of requests made by universities in the Marmara Region is 99,252 (54%), the number of requests that came to universities in the Marmara Region is 96,079 (53%). In total, 68% of the requests to the Marmara Region are from the universities in the same region, and the Marmara Region makes the majority of the requests (66%) from the universities in the same region. In addition to the intense interaction within itself, which corresponds to almost one-fifth (18%) of the total interaction in KITS, the Marmara Region receives requests from all other regions, and

Figure 2 Interregional cooperation network based on KITS data



Source: By authors' Çölden Akgül & Doğan

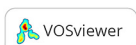
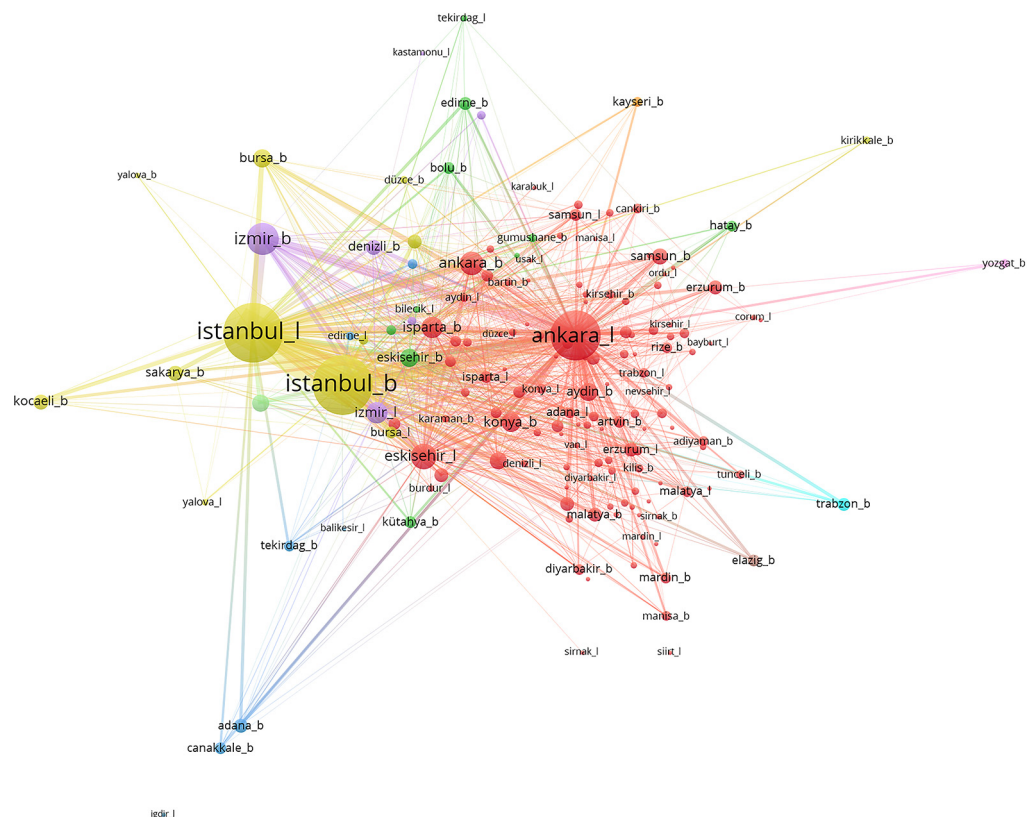
the Marmara Region makes requests to all other regions. The region in which the Marmara Region requests the most resources after itself is the Central Anatolia Region (26,318 requests; 26.5%). After the Marmara Region, the most active region in the network is the Central Anatolia Region. This region considerably differs from the Marmara Region. The universities in the Central Anatolian Region received 65,826 resource requests (36%), while they made a total of 19,050 requests themselves (10.5%). The region's central location in Türkiye is considered to be a significant factor in the high volume of requests it received. The fact that all regions except the Marmara and Aegean Regions are included in the red cluster, which includes the Central Anatolia Region, is also an indicator of this.

Figure 3 shows cooperation for ILL at the level of cities. The network is based on the resources requested (-l, lender) from each of 80 cities and the resources each city requests (-b, borrower). There are seven clusters with more than one city in the network. The clusters containing only one city each (Yozgat_b, Elazığ_b, Kayseri_b, Trabzon_b) are also located far from the network. This is because although these four cities requested between 510 and 1,282 resources from other cities, they only received 16–35 requests themselves. A similar situation exists for all cities located further out of the network. The most crowded cluster is the red cluster with 120 nodes. In

total, 37% of the total transactions are realized in the red cluster. The most prominent node in this cluster is Ankara, where 54,762 resources are requested from all 80 cities (30% of the total resource requested). İstanbul comes first among the cities that request the most resources (17,403, 32%) from universities in Ankara, followed by İzmir (6,113, 11%). Although Ankara receives a significant number of requests and plays a visible role in the network, it does not equally dominate in terms of requesting resources from other locations. The reason for this is that Ankara, where 30% of the total resource requests come from, makes only 3.5% of the total requests. Ankara has requested a total of 6,460 resources from 66 different cities.

Although it contains very few nodes compared to the red cluster, the most transactions in the network occurred in the yellow cluster (178,313, 49%). Almost all of these transactions (174,619; 98%) were made by İstanbul. Approximately half of the total requests come from (47.3%; 86,136) or are sent to (48.6%; 88,483) İstanbul universities. Contrary to Ankara, 62% of the total requests to universities in İstanbul were made mostly from İstanbul (54,665). The second city that requested the most from İstanbul is İzmir (6,089; 7%). There are nine nodes in the purple cluster, which is the third most prominent cluster in the network (28,584 transactions; 8% of the total transactions), the most visible of these are the ones belonging to

Figure 3 Intercity cooperation network based on KITS data



Source: By authors' Çölden Akgül & Doğan

İzmir. Unlike Ankara, İzmir is a city that requests more than it provides. The number of requests made by İzmir (15,788) is approximately three times higher than the resources requested from İzmir (5,569).

Figure 4 shows 182,033 resource requests in terms of universities. There are 161 universities, whose number of transactions varies from 1 to 32,972. Since each university has two different roles (-l, resource requested; -b, resource requester), there are 322 nodes on the network. In total, 82% of all requests came to 27 universities, whereas these universities requested 53% of the total. In total, 156 different universities requested resources from Boğaziçi University, which is very prominent in the orange cluster. Almost one-fifth of all resources are requested from this university. Other universities in Istanbul are among the universities that request the most from Boğaziçi University, and all are foundation universities. Bilkent University, which stands out in the red cluster, received requests from 154 different universities, corresponding to 13% of the total requests. Another prominent university in the network is METU. Although Boğaziçi, Bilkent and METU received 40% of the total resource requests, they made only 2.3% of the requests to other universities. In contrast, Kadir Has, Süleyman Demirel and Yaşar universities stand out for their higher number of submitted requests to other universities

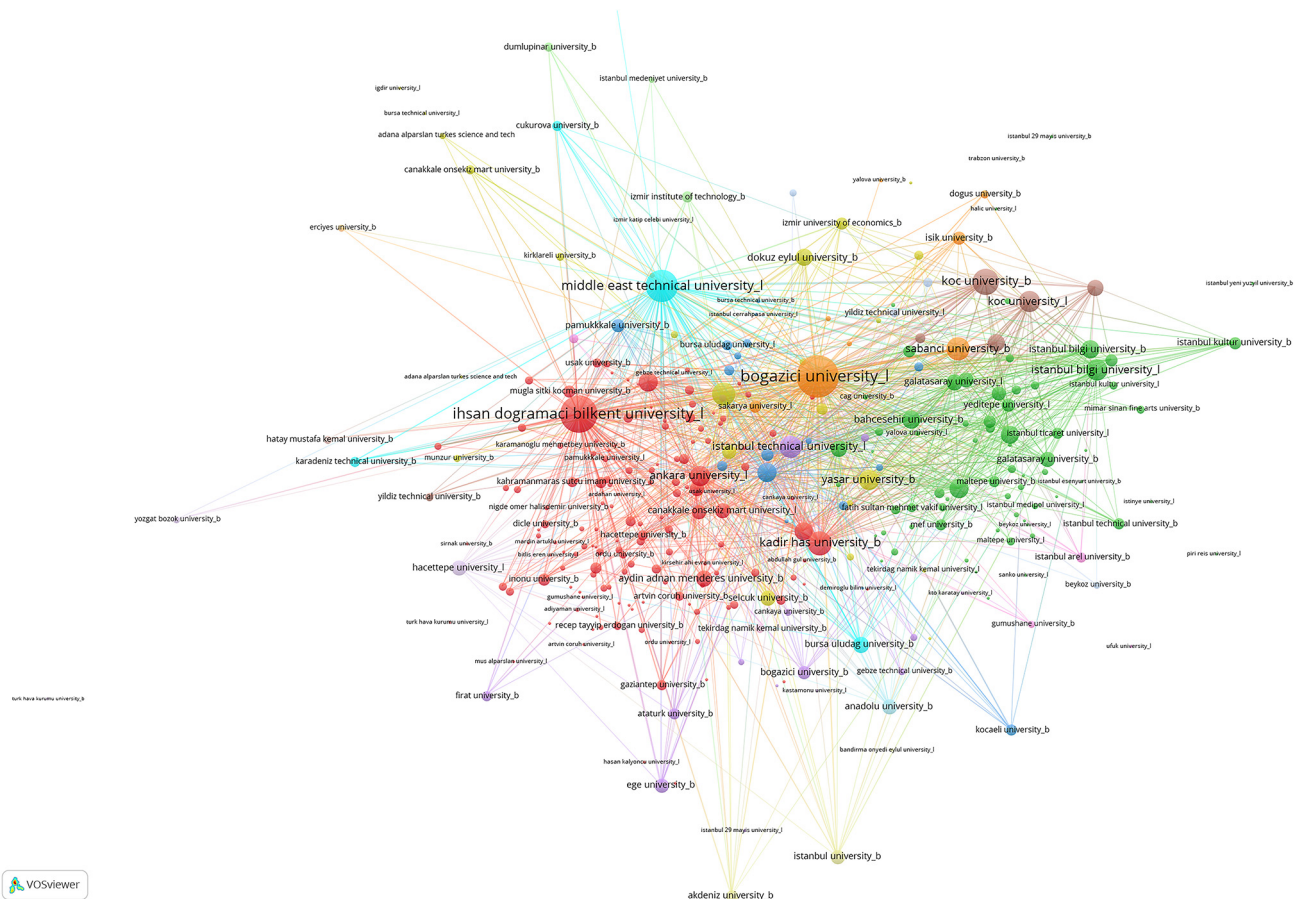
compared to the requests they receive. Similarly, Koç University, another prominent university in the network, follows this pattern.

4.2 Thematic distribution

The process of determining the subjects of the requested resources was carried out in two stages. In the first stage, main and subheadings were determined based on the call numbers. In the second stage, the subjects were determined based on the titles. The main and subheadings of 142,404 printed books, which constitute 78% of the total requests, could be determined. This data shows that the majority of resources shared through KITS are in print format. As also presented in Table 2, the most common headings for resources requested are Social Sciences, Language and Literature, World History and History of Europe, Asia, Africa, Australia, New Zealand, etc., and Law, and these headings represent almost half of the total requests (82,969; 46%).

Figure 5 shows the subheadings with more than 500 requests together with the main headings. The subtopic that stands out in the figure is History (General) under the main topic of World History. In total, 84% of the requests in the World History main heading are in the History (General) subheading. The other two subheadings with the highest requests, Europe and Law in General, Comparative and Uniform Law and Jurisprudence,

Figure 4 Interuniversity cooperation network based on KITS data



Source: By authors' Çölden Akgül & Doğan

Table 2 Main headings of requested printed books

Main headings	No. of subheadings	No. of requests	% of requests Total	% of requests General total
Social Sciences	16	27,770	19.5	15.3
Language and Literature	19	22,294	15.7	12.2
World History and History of Europe, Asia, Africa, Australia, New Zealand, etc.	14	17,103	12.0	9.4
Law	10	15,802	11.1	8.7
Philosophy, Psychology, Religion	15	10,588	7.4	5.8
Fine Arts	8	9,072	6.4	5.0
Political Science	13	8,081	5.7	4.4
Science	12	7,446	5.2	4.1
Technology	17	5,988	4.2	3.3
Education	9	4,452	3.1	2.4
Geography, Anthropology; Recreation	10	4,202	3.0	2.3
Medicine	16	2,863	2.0	1.6
Music and Books on Music	3	1,523	1.1	0.8
Auxiliary Sciences of History	10	1,454	1.0	0.8
History of the Americas	1	893	0.6	0.5
General Works	8	821	0.6	0.5
Bibliography, Library Science, Information Resources (General)	2	788	0.6	0.4
Agriculture	6	586	0.4	0.3
Military Science	9	549	0.4	0.3
Naval Science	4	129	0.1	0.1
Total	202	142,404	100.0	78.2
General Total	238	182,033	–	100.0

Source: By authors' Çölden Akgül and Doğan

both are under Law main headings. In total, 87% of the requests (13,814) in the main headings of law are in these two subheadings. Out of the 16 subheadings under the main heading of Social Sciences, Industries, Land Use and Labor and Commerce came to the fore with 35% of the requests. Literature (General), Philology, Linguistics and English Literature are the most requested subheadings in the main heading of Language and Literature (13,013; 58%). In addition to these, Architecture (4,223, 46%) subheading under Fine Arts, Mathematics sub-heading under Science (3,775, 51%) and Psychology sub-heading under the Philosophy, Psychology, Religion (3,073; 29%) main heading are other subtopics that stand out in terms of the requests.

The network shown in Figure 6 was created based on the words/phrases automatically drawn with VOSviewer from the titles of 181,495 out of 182,033 resources requested. Five clusters stand out in the network, the findings obtained from the network support the results for the main and subheadings. The red cluster (347 nodes) shows the words/phrases that are most frequently used in the titles of the requested resources in the fields of history, the blue cluster (337 nodes) represents the language and literature, and the yellow cluster (287 nodes) represents law. The green cluster (338 nodes) mostly focuses on analysis methods and applications, and in this sense, psychology and education fields with more applied studies come to the fore. In the purple cluster (194 nodes), the priority issues are economics and politics.

4.3 Factor affecting resource sharing

The requests of 107 state universities (53.5%) and 54 foundation universities (46.5%) out of a total of 161 universities are close to

each other. On the other hand, the requests received from state universities (66.1%) are almost twice that of foundation universities (33.9%). The median number of requested resources from universities does not differ statistically significantly regarding university type ($U = 2,847,500$; $z = -0.149$; $p = 0.882$). The result is the same for the median number of requested resources by universities, either ($U = 2,605,500$; $z = -1.015$; $p = 0.310$).

Based on the idea that the universities from which resources are requested more are established and, therefore, have a wider collection, and the universities that request more are younger universities, which may have more limited collections, it has been investigated whether the number of requests is related to the age of the university. A statistically significant correlation was found between the number of resources requested from universities and the ages of universities, although not strong ($r_s = 0.690$; $p < 0.001$; $R^2 = 0.48$). For about half of the universities, it can be said that as the age of the university increases, the number of resources requested from them also increases. On the other hand, the number of resources requested by universities is related to the ages of the universities for only 36% of universities ($r_s = 0.605$; $p < 0.001$; $R^2 = 0.36$).

To further analyze the impact of university age on their requests and the resources requested from them, we divided the universities into two groups: those established after 2000 (between 2000 and 2019) and those established before (20 years or older). We compared the medians of these two groups. In total, 91 universities established after 2000 benefit from KITS. The median number of resources requested by the universities younger than 20 and by the universities 20 or older differ statistically significantly ($U = 1,226,500$; $z = -6.679$; $p < 0.001$; $r_g = 0.615$). There is also a significant difference

The chart displays the following data series (estimated from the image):

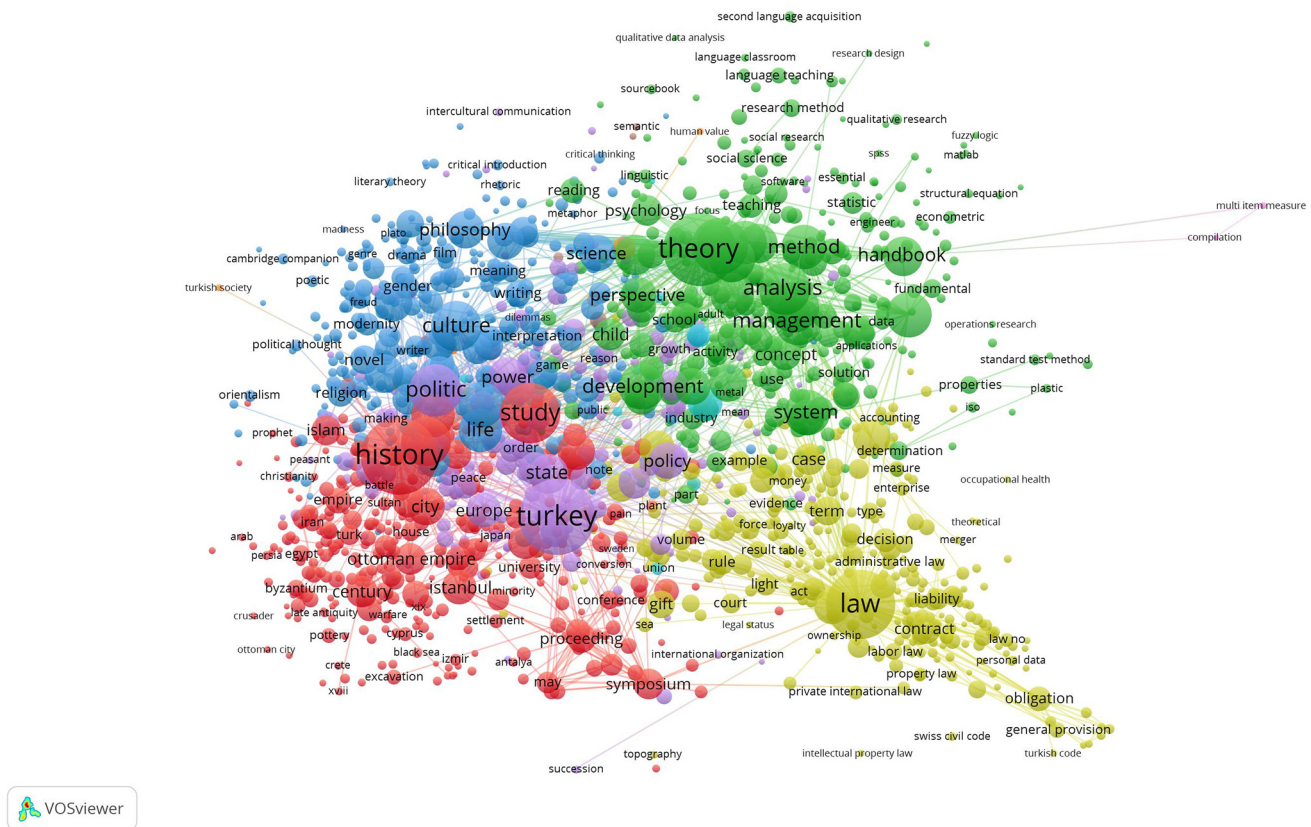
Category	Count
Literature (General)	4726
Language and Literatures of Eastern Asia, ...	1649
American Literature	1338
English Literature	792
French Literature	8370
Italian Literature - Italian Literature - Spanish, ...	5444
Law in General, Comparative and Unitof...	649
History (General)	14,395
Psychology	3073
Philosophy (General)	2825
Religions, Mythology, Rationalism	1300
Islam, Bahaism, Theosophy, etc.	1114
Speculative Philosophy	648
Architecture	4223
Visual Arts	1787
Decorative Arts	1147
Painting	723
Sculpture	508
Political Theory	1735
General Legislative and Executive Papers	1161
International Relations	1083
Political Institutions and Public Administration ...	771
Political Science (General)	661
Political Institutions and Public Administration ...	559
Science	935
Mathematics	3775
Physics	866
Theory and Practice of Education	2581
Special Aspects of Education	642
Geography (General), Atlases, Maps	1346
Geography, A...	872
Manners And Customs (General)	576
Recreation	545
Engineering (General), Civil Engineering	1305
Technology (General)	749
Manufactures	600
Electrical Engineering, Electronics, Nuclear ...	575
Medicine (General)	1378
Internal Medicine	656
Music	798
Music On Music	597
Greece	1186
History of the Americas	893
Books (General), Writing, Paleography, Book In ...	779
Bibliography, ...	779
World History, ...	597
History of L...	597
Music and B...	597
Medicine	597
Industries, Land Use, Labor	4959
Commerce	2882
Sociology (General)	2882
The Family, Marriage, Women	2178
Economic Theory, Demography	2178
Economic Sciences (General)	2178
Social Pathology, Social and Public Welfare...	1895
Communities, Classes, Races	1434
Finance	1323
Social History and Conditions, Social Proble...	726

between universities in these two age groups regarding the number of resources requested from universities ($U = 844,000$; $z = -7,983$; $p < 0.001$; $r_G = 0.735$).

The size of the university library collections and the number of library users can be effective in resource requests. For this reason, it was investigated whether there was a statistical

correlation between the size of the collection and the number of resource requests, and no significant correlation was found (between the number of resources requested from universities and the size of their collections: $r_s = 0.493$; $p < 0.001$; $R^2 = 0.24$, and between the number of resources requested by universities and the size of the collection: $r_s = 0.394$; $p < 0.001$;

Figure 6 Network created according to the titles of the resources requested



Source: By authors' Çölden Akgül & Doğan

$R^2 = 0.16$). Using the Spearman correlation test, it was investigated whether the number of resources requested by the university libraries and the resource requests received through these systems were related to the number of library users. Contrary to what was found for the collection size, a significant correlation was found for both ($r_s = 0.468$, $p < 0.001$, $R^2 = 0.22$; and $r_s = 0.356$, $p < 0.001$, $R^2 = 0.13$, respectively).

5. Discussion

In this study, KITS data were primarily evaluated geographically as in similar studies in the literature (McGaugh, 1994; Xing *et al.*, 2020). Geographical evaluation of the KITS data showed that the Marmara Region stands out in ILL. The population, the economic activity and the number of universities are high in Marmara Region, and the most populous city of Türkiye (İstanbul) is in this region, all of which should be considered when evaluating this result. Since other cities with large populations are located in Central Anatolia and the Aegean Region, these regions lead cooperation in resource sharing. When the cooperation between cities is evaluated at the city level, resource sharing is concentrated in three big cities (İstanbul, Ankara and İzmir). This situation is also related to the high number of universities when evaluated in terms of both population and economy, as in the Marmara Region.

An interesting finding was the limited cooperation among universities within Ankara. The observed trend may be explained by the establishment of reciprocal agreements among Ankara universities, which facilitates access to resources in other university libraries in the city for their respective users, without necessitating ILLs. The geographical analysis conducted in this study also serves to identify potential cities for establishing reciprocal agreements. The results indicate that İstanbul and İzmir, with their numerous universities, emerge as promising candidates for such agreements. Based on the findings of this study, the establishment of regional ILL systems, in addition to central systems like KITS, holds potential benefits for specific areas. Considering the results, it is suggested that three regionals ILL systems could be implemented. First, an efficient regional system for the Marmara Region would be recommended. Additionally, a system dedicated to the Aegean Region has the potential to be effective. Finally, another regional system could be considered for the remaining five regions. These regional ILL systems have the potential to enhance resource sharing and improve access to materials within their respective geographical areas.

The analysis of the Interlibrary Loan Tracking System (KITS) revealed that resource sharing predominantly occurs in the field of "Social Sciences." The most requested subjects include "Language and Literature," "World History," "Law,"

“Philosophy and Psychology” and “Fine Arts.” These findings reflect the information-seeking behaviors in academic research in Türkiye and align with similar studies in the literature (Cuhadar *et al.*, 2019; Güran and Kaya, 2017; Porat, 2003; Teper *et al.*, 2017). Institutions can leverage this information to enhance their collections through collaborative efforts. Sharing this thematic analysis with librarians responsible for collection development provides valuable insights into user research interests and information needs, enabling them to make informed decisions regarding resource acquisition. Furthermore, it encourages collaboration among libraries based on shared thematic interests, thus enhancing the efficiency of the ILL system.

By implementing a centralized system to track data, institutions can gather comprehensive statistical information and offer cost-effective lending options and on-site access for users. This not only enhances resource sharing but also allows for efficient collection development. Additionally, it enables institutions to identify areas where collaboration and reciprocal agreements can be established, benefiting both resource providers and recipients. To ensure up-to-date knowledge on system usage and internal operations, institutions should conduct similar studies using their own data, both at a regional and thematic level. This will provide them with valuable insights into the specific needs and requests of their user community, allowing for targeted improvements and informed decision-making. By continually analyzing and refining their ILL practices, institutions can better serve their users and strengthen the overall effectiveness of resource sharing networks.

Resource sharing initially focused on print materials, as evident from the KITS data. The system was primarily designed for printed source submission, with books being the most commonly shared resources. However, with technological advancements, electronic resource sharing in the form of PDFs became more prevalent over time. Nonetheless, some institutions still opt for print delivery due to licensing agreements, which hinders quick access to resources. To facilitate the expansion of e-resource sharing and address these challenges, it is recommended to implement a regulated lending system and promote collaboration based on sharing rights. Additionally, incorporating ILL sharing clauses in e-resource contracts, if they are not already present, is advisable. These measures aim to improve access to resources, enhance the efficiency of resource sharing networks, and positively impact the library budget.

Upon analysis of the number of requests received and made by various types of universities, it has become evident that foundation universities are more inclined to submit requests, whereas state universities tend to receive a greater volume of requests. This phenomenon may be attributed to the relatively longer history of state universities in Türkiye, compared to the more recent establishment of a majority of foundation universities. For librarians at foundation universities, addressing collection gaps is crucial and analyzing and evaluating ILL data can help finding these gaps and efficiently manage and develop their collections. State universities should allocate adequate personnel to handle the increased requests. Overall, strategic resource management and collaboration are essential for both types of institutions.

University age can be considered an important factor in resource sharing. As a matter of fact, for half of the universities, the number of requests increases as the age increases, and more

resources are requested from older and, therefore, more well-established universities. Librarians at both types of institutions can use this information to optimize their resource sharing practices and allocation of resources. For older universities, they can plan for higher requests and ensure that they have adequate resources to fill the requests. For newer universities, they may need to work on building their collection and resource sharing partnerships to meet the needs of their users. The expected result could not be achieved in terms of the effect of library users and collection size on resource sharing. The potential reason for this is that the user groups that make requests are only researchers. In addition, this result may also be related to the amount of time universities use the KITS system, depending on the age of the institution. It is thought that the quality of the collection is considered when making requests, not the size of the collection. The conclusion of the current study differs from that of a previous one, which found that university libraries with larger collections receive and send more requests than those with smaller collections (Porat, 2003).

After analyzing the KITS data, it is clear that this shared system effectively supports the resource sharing of printed materials among university libraries in Türkiye. When the most remarkable findings from the KITS data are examined, it has been revealed that the cooperations are made effectively throughout Türkiye, the institutions that request resources are diverse, and the institutions that are in the first place send resources mostly. It is foreseen that the requested institutions have a heavy workload and that the institutions should support this process by increasing the number of personnel. It is important to support this process by enabling the requesting institutions to diversify the institutions for which they want resources. As can be understood from the emerging networks, it is a remarkable result that resource exchange is more intense in populated areas and that cooperation with institutions in neighboring regions and cities is more. Regional collaborations and restructuring of the system can provide a better analysis of both the process and the data. By carrying out different collaborations of institutions through the system, both statistical and on-site use can be achieved. Another important finding is that, with the thematic analysis of the research, institutions contribute to the institution's budget by developing their own collections.

Furthermore, encouraging e-resource sharing will contribute to the institution's budget. As a result, if member libraries adhere to the recommendations in this article, they can further strengthen interinstitutional collaboration and advance resource sharing within Türkiye.

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Appendix

Table A1 Data in the three KITS data sets obtained and the ones used (in italics) in the study

Data set 1: Books	Data set 2: Articles	Data set 3: Thesis and book chapters
<i>Request number</i>	<i>Request number</i>	<i>Request number</i>
<i>Title</i>	<i>Title</i>	<i>Title</i>
–	–	Type
Author	Author	Author
Publication info	–	Publication info
–	–	Date
<i>Call number</i>		Call number
ISBN	ISSN	ISBN
–	–	Chapter title
–	–	Author
–	–	Pages
Request date	Request date	Request date
Return date	–	–
<i>Lender (Requested university)</i>	<i>Lender (Requested university)</i>	<i>Lender (Requested university)</i>
<i>Borrower (Requesting university)</i>	<i>Borrower (Requesting university)</i>	<i>Borrower (Requesting university)</i>
States of process steps (0–14)	States of process steps (0–10)	States of process steps (0–14)
0 = Record deleted	0 = Record deleted	0 = Record deleted
1 = New request	1 = New request	1 = New request
2 = Request reached library	2 = Request seen	2 = Request seen
3 = Accomplished and mailed	3 = Accomplished and mailed	3 = Accomplished and mailed
4 = The requested book reached the library	4 = Request reached requested library	4 = Accomplished, document uploaded
5 = Book borrowed to user	5 = Request sent electronically	5 = Request reached requested library
6 = Extension requested	6 = Request downloaded electronically	6 = Request not reached requested library
7 = Extension request accepted	7 = Request not fulfilled	7 = Request downloaded
8 = Extension request denied	8 = Request not fulfilled, seen	8 = File deleted
9 = Recalled	9 = Request closed	–
10 = Returned and mailed back	10 = Notes	–
11 = Request not fulfilled	–	11 = Request not fulfilled
12 = Request not fulfilled, seen	–	12 = Request not fulfilled/ Deleted, seen
13 = Resource returned and request closed	–	13 = Request closed
14 = Note left	–	14 = Note left
Source: By authors' Çölden Akgül and Doğan		

Appendix 2

Table A2 LC main headings corresponding to DDC main headings and their distribution in the data set

No	DDC	LC	LC main heading	Total
1	300	H	Social Sciences	27,770
2	400–800	P	Language and Literature	22,294
3	900	D	World History and History of Europe, Asia, Africa, Australia, New Zealand, etc.	17,103
4	340	K	Law	15,802
5	100–200	B	Philosophy, Psychology, Religion	10,588
6	700	N	Fine Arts	9,072
7	320	J	Political Science	8,081
8	500	Q	Science	7,446
9	600	T	Technology	5,988
10	370	L	Education	4,452
11	910	G	Geography, Anthropology, Recreation	4,202
12	610	R	Medicine	2,863
13	920	C	Auxiliary Sciences of History	1,454
14	780	M	Music	1,523
15	970–980	E-F	History of the Americas	893
16	010–020	Z	Bibliography, Library Science, Information Resources (General)	788
17	000	A	General Works	821
18	630	S	Agriculture	586
19	355	U	Military Science for Military History, see D–F	549
20	359	V	Naval Science for Naval History, see D–F	129
	Total			142,404

Source: By authors' Çölden Akgül and Doğan

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